Tanish Srikakolapu

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Education

University of Waterloo **2024 - 2029**
Bachelor of Mathematics in Combinatorics and Optimization *Waterloo, ON*
Relevant Coursework: Data Structures and Algorithms, Probability, Computational Mathematics

Experience

AI Solutions Intern **Apr 2026 – Present**
GuruvaiSciences *Remote*

- Improved inference-model selection by evaluating candidate models on structured datasets across **4 performance dimensions**: output quality, consistency, execution performance, and failure handling.
- Increased ML agent safety across **3 high-risk validation areas** by designing prompts and guardrails that detected data leakage, invalid splitting strategies, and unsupported modelling assumptions.
- Improved end-to-end inference reliability across **3 agent stages** by tracing LangGraph execution paths, debugging model failures, and correcting routing between planning, profiling, and execution components.

Software Developer Intern **Feb 2025 – Present**
Electrium Mobility *Waterloo, ON*

- Engineered a **BLE scanning system** using **Android background services** to enable persistent real-time tracking across **15+ nearby devices**, reducing manual discovery overhead by **40%**.
- Built a **real-time Android dashboard** in **Kotlin** displaying live device status and connectivity metrics for **3+ e-bike fleets**, improving operator visibility and response time by **30%**.
- Maintained **99%+ BLE connection reliability** across app state changes by integrating scanning with **Android's background service lifecycle**, reducing dropped connections by **25%**.

SWE Instructor **Sep 2025 – Dec 2025**
Exceed Robotics *Richmond Hill, ON*

- Instructed **30+ students** aged 6–15 across **40 lessons** in **programming and robotics**, increasing hands-on technical exposure across multiple skill levels.
- Introduced **computer vision concepts** via **OpenCV and Google Cloud Vision API** across **5 project-based modules**, bridging theory and real-world application for beginner programmers.
- Delivered **40+ lessons** across **Arduino, Python, LEGO, VEX, Raspberry Pi**, tailored to students' age and skill.

Projects

Nautica | Python

- Designed and implemented an **AI agent** for a **12-hour Battleship competition** using probabilistic modeling and search heuristics.
- Applied **Monte Carlo simulation** to estimate ship placement probabilities and optimize move selection dynamically.
- Achieved a **69% win rate**, earning **1st Place (Round Robin)** and **Most Creative Implementation** at **Code Clash UofT (2026)**.

Clippy | JavaScript, HTML, Node.js

- Built a **full-stack web application** using JavaScript, HTML, and Node.js with a responsive UI.
- Integrated the **OpenAI API** to generate AI-assisted drawing prompts.
- Awarded **2nd Place** at the **MUES Hackathon (Oct 2025)**.

Discord Meeting Notes Generator Bot | Python, Discord.py, OpenAI API

- Developed a **Discord bot** in **Python** to **record voice calls** and generate **automated meeting summaries**.
- Implemented **speech-to-text** using **Opus** and **Whisper**.
- Generated **participant-specific action items** using **LLM-based summarization**.

Technical Skills

- **Languages:** Python, Java, C, C#, JavaScript, TypeScript, SQL, Kotlin, HTML/CSS
- **Technologies:** React, Node.js, Flask, FastAPI, Streamlit, SQLite, Supabase, PostgreSQL
- **Developer Tools:** Git, Docker, Linux, VS Code, Android Studio, Jupyter Notebook
- **Cloud & APIs:** AWS Lambda, OpenAI API, REST APIs, Bluetooth Low Energy (BLE)